

Boardgames Are Made of Rules

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Abstract

This report checks how well local Large Language Models can understand, analyze and create board game rules.

I created a five-task test to evaluate them: explaining rules, analyzing properties, correcting rules, suggesting moves, inventing games

I used this test on five board games with three models: Llama 3.2, Mistral Nemo, Qwen 2.5

All these models were run locally using Ollama. I also used a model, Claude Opus on some tasks to see how well the local models did compared to a high-end model. The results show that local models can change their tone to fit audiences. The task of suggesting moves showed a difference in ability, between models. Mistral did a job with the game Pandemic but smaller models suggested moves that are not allowed. When it came to inventing games local models all created games that were similar. They all used hex grids. Had players try to get points. A commercial model, Claude Opus was able to create a varied game.

1 Introduction

LLMs have provided impressive results in many tasks of NLP , including translating, summarizing, writing code and answering questions. Recently, some researchers have begun to explore whether LLMs are able to apply their abilities to develop structured reasoning, which requires following, changing and developing systems governed by rules. The use of board games provides an excellent platform to evaluate this theory. The rulebook contains a complete formal system defining how to setup the game, how to take turns, what moves can be made and how winners and edge cases are defined. One must read the rules logically to comprehend them, develop a step-by-step simulation of the rules, and then create new items given the restrictions.

I present a five-task evaluation pipeline that tests LLMs across the full lifecycle of board game rules: (1) **explaining** rules to different audiences, (2) **analysing** game properties, (3) **correcting** flawed or ambiguous rules, (4) **recommending moves** from intermediate game states, and (5) **inventing** entirely new games.

2 Methodology

2.1 Research Questions

I decompose the problem into five sub-topic. Each topic must follow a question:

- RQ1** Can LLMs produce accurate, audience-adapted explanations of board game rules?
- RQ2** Can LLMs extract structured game properties (mechanics, complexity, player count, duration) from natural-language rulebooks?
- RQ3** Can LLMs identify ambiguities, inconsistencies, or missing rules in existing rulebooks?
- RQ4** Given a game state described in natural language, can LLMs recommend valid and strategically sound moves?
- RQ5** Can LLMs generate coherent, playable rule systems for entirely new board games?

2.2 Pipeline Architecture

I implement a modular pipeline following the Template Method design pattern (Figure 1). Each pipeline inherits from an abstract base class defining five stages: (1) a data adapter loads the input, (2) a text cleaner normalises formatting, (3) a task-specific prompt strategy constructs the LLM input, (4) the Ollama client generates the response, and (5) a parser extracts and compose the output. All five tasks share this skeleton but specialise prompt construction and output parsing.

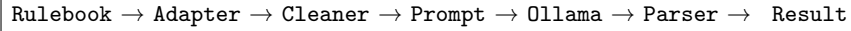


Fig. 1 Pipeline execution flow.

2.3 Models

I evaluate three open-weight models served locally via Ollama, selected to represent different model families and parameter scales:

Table 1 Models used in the experiments.

Model	Alias	Parameters	Context	Temperature
Llama 3.2 [1]	llama	3B	8K	0.7
Mistral Nemo [2]	mistral	12B	128K	0.8
Qwen 2.5 [3]	qwen	7B	32K	0.6

2.4 Dataset

I select five board games spanning a range of complexity, mechanics diversity, and release dates:

Table 2 Board games used in the experiments, with BGG reference properties.

Game	BGG Weight	Players	Duration	Primary Mechanics
Century: Spice Road	1.80	2–5	45 min	Set Collection, Hand Management
Cascadia	1.84	1–4	45 min	Tile Placement, Pattern Building
Pandemic	2.39	2–4	45 min	Cooperative, Hand Management
Ticket to Ride	1.82	2–5	60 min	Route Building, Set Collection
Wingspan	2.48	1–5	70 min	Engine Building, Card Drafting

2.5 Evaluation Metrics

Explain task. In order to assess the explanation provided, I recorded the three automated metric values of three cover metrics: (1) coverage, the proportion of Key Game Terminology that is contained within Explanation (2) score based upon distance of average sentence length (3) length ratio value of how closely the number of words in Explanation is to the target.

Analyse task. Mechanics are evaluated using set-based F1 after normalisation via a custom LLM-to-BGG mapping (around 85.% of the terms that LLM generated was successfully mapped). Complexity accuracy is the absolute difference from the BGG value, scaled to a 0–1 range. Player count and duration are evaluated as exact match and percentage error.

Qualitative tasks. Fix-rules, move, and invent are evaluated through manual inspection of structural completeness, logical consistency, rule adherence, and factual accuracy. For invent, I additionally test whether generated games pass automated schema validation.

3 Experimental Results

I conducted 121 experiments across four models: 60 for explain (5 games $\times$ 3 audiences $\times$ 4 models), 20 for analyse, 10 each for fix-rules and recommend-move (5 games $\times$ 2 model groups), and 21 for invent (15 local + 6 Claude). Every output was inspected manually in addition to automated scoring.

3.1 Task 1: Rule Explanation

All three local models show clear audience-dependent scaling: expert explanations are longer than child-level ones (Table 3).

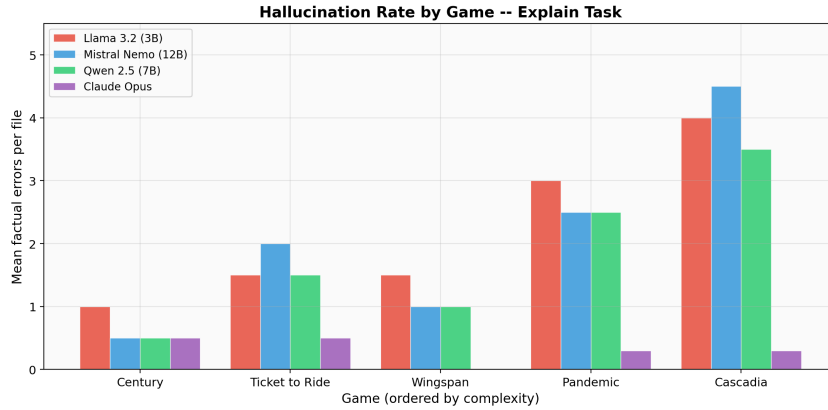
I can see that the expert explanations build the strategy component based on fabricated rules, the child-level simplifications at the same time may be adding a

Table 3 Average token count by model and audience level.

Model	Child	Adult	Expert
Llama 3.2	446	667	849
Mistral Nemo	567	767	942
Qwen 2.5	550	935	1014
Claude Opus	806	1002	1731

structural issue. A case in point is Llama reflecting Cooperative Pandemic as Competitive, whereas Qwen creates an imaginary card cost associated with curing the diseases where one does not exist.

The biggest similarity is that none of the local models will acknowledge uncertainty in any way. When a rule is not cited, they will create it, but the results will be consistent with each other; Llama creates mechanics with a thematic nature; Mistral creates numbers and calculations; Qwen creates plausible data that is not accurate.

**Fig. 2** (Explain task). Error rates correlate with game recency rather than complexity.

3.2 Task 2: Game Property Analysis

Table 4 Game property analysis: mechanics F1 and overall accuracy against BGG ground truth.

Model	Mech. F1	Player Count Match	Overall
Llama	12.9	2/5	40.7
Mistral	4.3	2/5	45.8
Qwen	18.3	4/5	48.8

Identification weaknesses exist in the mechanics segment of the board game. The average F1 of all fifteen local-model tests is 11.8% (Claude at 54.8%, see Section 5). Models produce generic labels instead of Board Game Geek terminology. Worker Placement has become the Hallucination Default for the Llama; it’s produced in four out of five games, including both Century and Pandemic—neither of which have any relation to it. BGG terminology (e.g., Action Points, Chaining, and Multi-Use Cards) is never identified by any model.

Llama and Qwen are overestimated by, on average, 1 points; Mistral by 0.60 on a 5-point scale. None of the models produce an underestimation. I think that they are more or less in a good range even if they are overestimating the difficulty. It seems that the verbosity of the rulebook may affect the output. In regards to the average player count, in exact matches, Qwen achieves 4 out of 5, Llama achieves 2 out of 5, and Mistral achieves 2 out of 5,

3.3 Task 3: Rule Correction

This task highlights how models can both correctly identify true ambiguities and propose false rules. Since there are many creative proposals applicable here, it will be more difficult to separate reasonable suggested fixes from those that have been hallucinated compared to other tasks.

The following two examples illustrate some errors. The first, a redundancy caused by not reading: Llama has proposed "the player who took the last turn in Century wins if there is a tie"; however, this rule exists in the text of the game rules. The model missed identifying an existing rule previously and re-proposed it. The second, an informed hallucination: Mistral invented 2 new types of special event cards in Pandemic, called "Direct Flight" and "Government Grant", each with made up mechanics; although the name of these events exists in Pandemic, the mechanics associated with them does not.

Mistral produces the most structured and functional proposed fixes across the 3 models. Llama produced the least functional fixes. Llama has the least confidence as measured by the use of [Insert corrected version...] in Pandemic response instead of providing a made-up replacement for the correct answer, which is in direct contrast to Mistral and Qwen who are totally confident about their made-up answers.

3.4 Task 4: Move Recommendation

This task requires multi-step reasoning over game state, rules, and strategy. It produces the largest performance gap between models.

Pandemic presents a dilemma: 3 red cubes in Bangkok (outbreak imminent) versus collecting the 5th blue card for a cure. Mistral produces a near-optimal three-action sequence: Treat Disease in Bangkok, Charter Flight to Paris, Share Knowledge, thus managing the crisis while progressing toward finding the cure. Llama recommends sharing knowledge with a Researcher it cannot reach and references "Jane in Chennai," a character and city from the rulebook’s illustrative example on page 8 rather than from the provided game state. Qwen ignores the red cubes entirely and misapplies the Medic’s special ability.

Cascadia is the most striking result. The game state explicitly says "You need 1 more bear adjacent to your current bear for the pair bonus," yet all three models fail. Llama and Mistral pursue long-term habitat expansion instead of the immediate scoring opportunity. Qwen selects the correct tile but places an elk instead of a bear, revealing a disconnection between high-level reasoning ("I need to complete the pair bonus") and concrete execution (choosing the wrong animal token).

3.5 Task 5: Game Invention

I tasked each model with inventing games from five thematic prompts (Silk Road, coral reef, wildfire, space routes, medieval scriptorium). All 15 experiments completed. Automated validation pass rates reflect structural quality: Mistral 4/5, Qwen 3/5, Llama 2/5.

The most revealing finding is the structural uniformity among local models. Despite five different themes, 14 of 15 locally generated games use a hexagonal grid, 12 include exactly 48 Resource Cards of 4 types, and all specify 2 to 4 players with sequential turns. 100% use victory points as the win condition. The models appear to follow an internalised "board game template" applied almost invariably, regardless of theme. No generated game features asymmetric roles, real-time elements, hidden information beyond secret missions, player elimination, or genuinely novel mechanics.

No model produces precise end-game scoring or addresses edge cases and no model reasons about balance actions.

4 Discussion

4.1 Cross-Task Summary

Table 5 summarises model performance across all five tasks, combining automated scores where available with qualitative ratings on a 0 to 10 scale for the remaining tasks.

Table 5 Cross-task performance summary.

Model	Explain	Analyse (%)	Fix	Move	Invent
Llama 3.2 (3B)	5.5	40.7	3.5	6.5	4.0
Mistral Nemo (12B)	6.5	45.8	7.0	7.5	7.0
Qwen 2.5 (7B)	6.0	48.8	6.0	5.0	5.0
Claude Opus (API)	8.0	54.8	8.5	9.5	8.5

4.2 Key Findings

Looking across all tasks, there is a clear split in what these models can and cannot do. Models do well on *linguistic* tasks: paraphrasing rules, adapting register, generating structurally coherent game descriptions. However, tasks requiring *precision* or

reasoning expose fundamental limitations: low F1, complexity overestimation, and the persistent refusal to admit uncertainty.

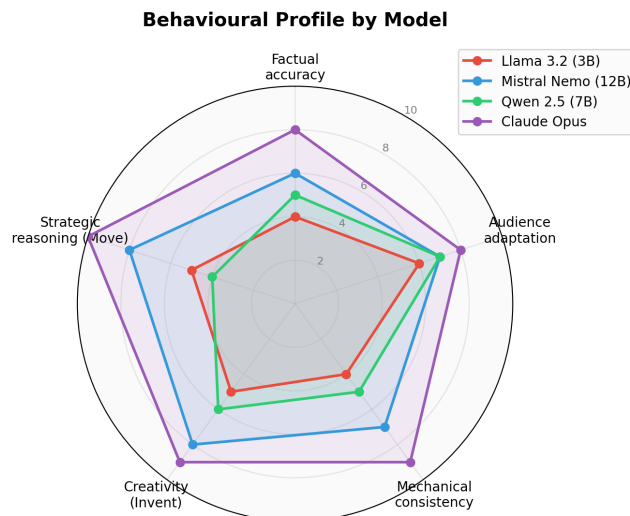


Fig. 3 Behavioural profile by model across five capability dimensions.

4.3 Comparison with a Commercial Model

To put the local model results in context, I ran all five tasks with Claude Opus via the Anthropic WebSite. The results are included in Table 5 and in the charts.

Explain. Claude produces longer, more detailed explanations than the local models, particularly for expert audiences. Factual errors drop to near zero: roughly 0.3 errors per file on average, compared to 1.5–3.5 for the local models. Claude also handles Cascadia well, where all local models struggle.

Analyse. Claude achieves a mechanics F1 of 54.8%, well above the local average of 11.8%. Its complexity predictions are close to BGG ground truth, and it matches player counts perfectly on all five games.

Fix-Rules. Claude identifies more possible issues. By contrast, local models produce 5-7 generated errors across the five games.

Move. This is the biggest gap. Claude scores 5/5 optimal moves, including the Cascadia bear-pair scenario that all local models missed.

Invent. Given the same five themes, Claude breaks the template convergence: only 3/6 of its games use hex grids (vs 15/15 local), and it introduces mechanics like route networks, push-your-luck, and market fluctuation that no local model ever produced. There are no new mechanics but the ideas of the games are a great mix of existing games.

4.4 Limitations

This work has some clear limitations. The fix-rules, move, and invent evaluations are primarily qualitative, as automated metrics for these tasks are difficult to define without human annotation. The game selection covers only light-to-medium complexity titles. Each experiment is a single run. BGG ground truth for mechanics is itself crowdsourced and occasionally inconsistent.

4.5 Future Work

There are a few obvious next steps: human evaluation by board game players for explanation quality and move validity; multi-turn game simulation to test rule adherence across an entire session; domain-specific fine-tuning on board game rulebooks and BGG data; and formal game simulators for automated consistency verification of generated rules.

AI Usage Disclaimer

This project was independently designed, developed, and implemented. All architectural decisions, pipeline design, and experimental analyses reflect original work.

I used Claude Opus 4, Anthropic in the final stages of the project for the following purposes: Code refinement: improving readability, consistency, and best practices in already-functioning code, Text polishing: reviewing and refining descriptive sections for clarity and fluency.

All AI-generated suggestions were reviewed, tested, and adapted before integration. No core logic, architectural design, or analytical reasoning was delegated to the model. The project structure, methodology, and interpretation of results are entirely my own.

I take full responsibility for the final content, its accuracy, and its academic integrity.

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References

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